Business Question and Visualization Report

Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	5 Marks

Visualization development refers to the process of creating graphical representations of data to facilitate understanding, analysis, and decision-making. The goal is to transform complex datasets into visual formats that are easy to interpret, enabling users to gain insights and make informed decisions. Visualization development involves selecting appropriate visual elements, designing layouts, and using interactive features to enhance the user experience. This process is commonly associated with data visualization tools and platforms, and it plays a crucial role in business intelligence, analytics, and reporting **Business Questions and Visualisation**

The process involves defining specific business questions to guide the creation of meaningful and actionable visualizations in Tableau. Well-framed questions help in identifying key metrics, selecting relevant data, and building visualisation that provide insights.

Sample

1. **What are the monthly sales trends?**
    - *Visualization*: Line chart showing monthly sales trends.
    - *Screenshot of visualisation*
2. **Which products have the highest revenue?**
    - *Visualization*: Bar chart comparing revenue by product ○ *Screenshot of visualisation*
3. **How is customer satisfaction varying across regions?**
    - *Visualization*: Map visualization showing customer satisfaction by region. ○ *Screenshot of visualisation*

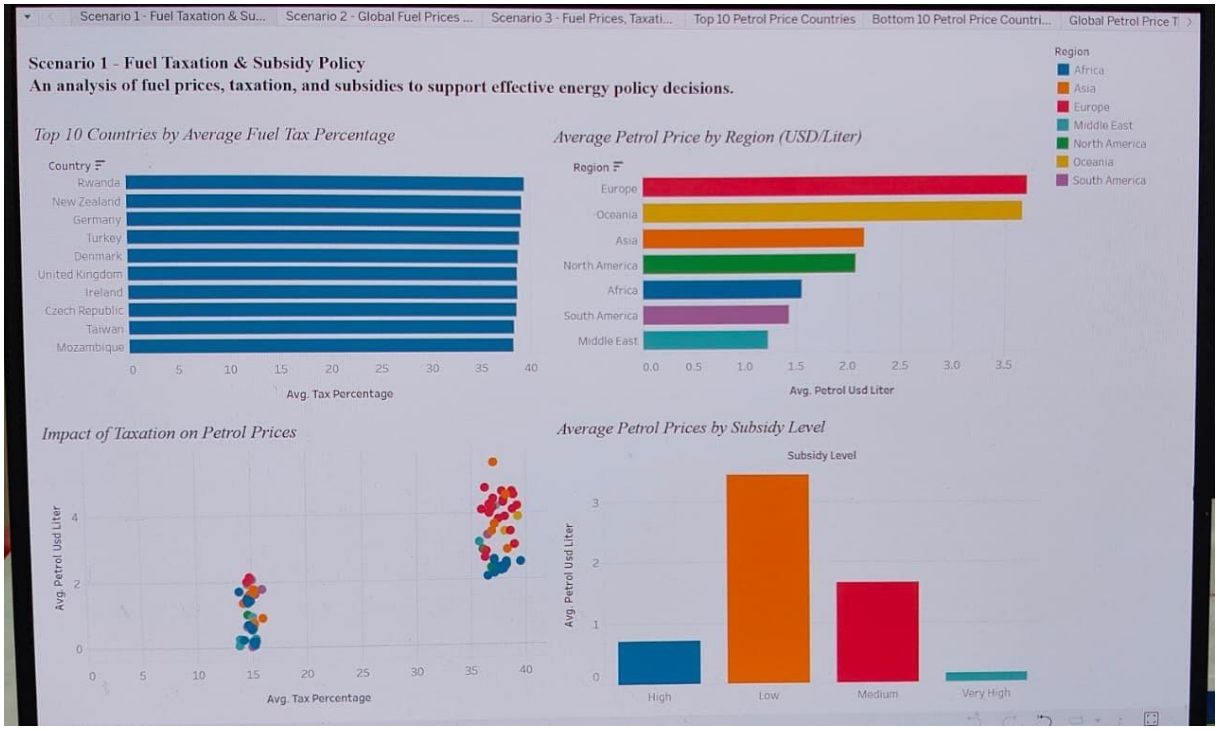
Note: Min 8 business question and visualisations Required inform of above

Project-Specific Business Questions and Visualisations

The following questions are addressed by the existing Tableau Public workbook and its three dashboards. The chart descriptions reflect the workbook structure; they are not claims of causal findings. Screenshots below are supplied as visual evidence of the dashboards available in the project materials.

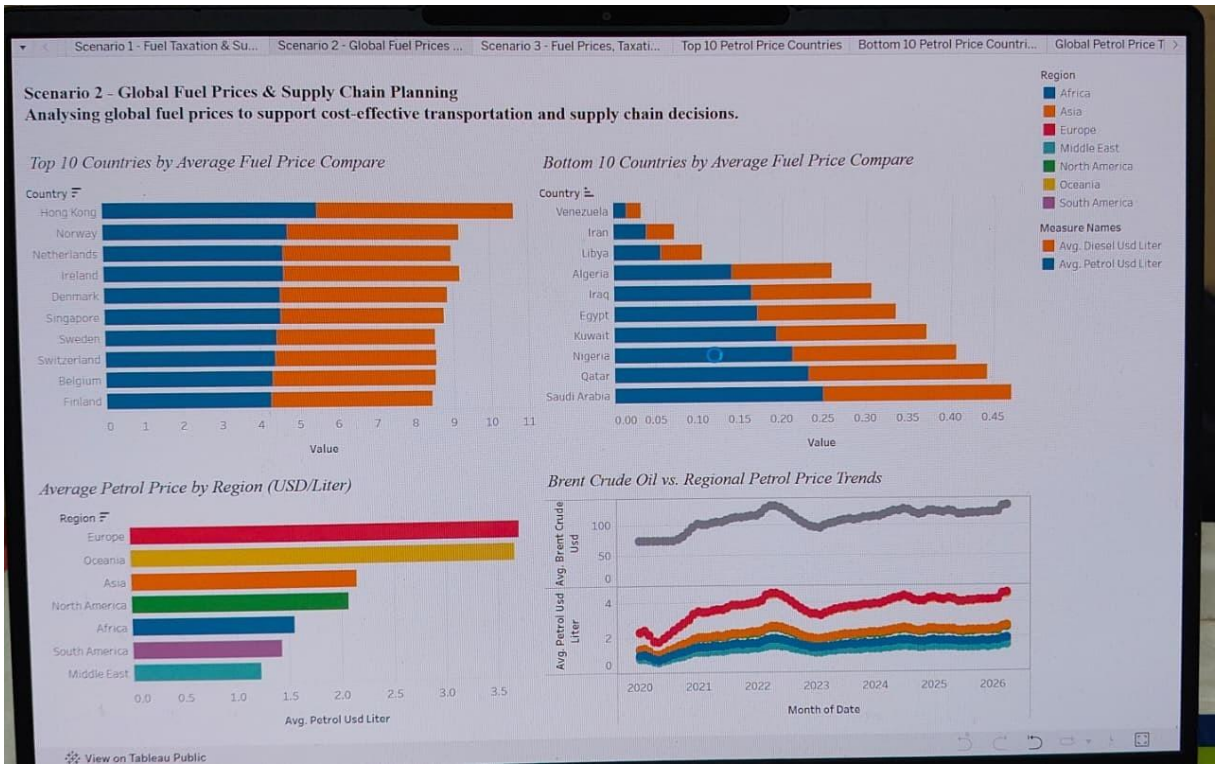
Business Question	Visualisation / Chart Type	Dashboard	Fields / Measures
1. Which countries have the highest average fuel tax percentages?	Horizontal bar chart: Top 10 Countries by Average Fuel Tax Percentage.	Scenario 1 — Fuel Taxation & Subsidy Policy	Country; average tax percentage.
2. How does average petrol price vary across regions?	Horizontal bar chart: Average Petrol Price by Region, with a Region legend.	Scenario 1 — Fuel Taxation & Subsidy Policy	Region; average petrol price (USD/liter).
3. What relationship can be explored between fuel taxation and petrol prices?	Scatter plot: Taxation vs Petrol Price. It supports comparison of association, not proof of causation.	Scenario 1 — Fuel Taxation & Subsidy Policy	Tax percentage; petrol price (USD/liter); country/region context.
4. How do average fuel prices differ by subsidy level?	Bar chart: Fuel Prices by Subsidy Level.	Scenario 1 — Fuel Taxation & Subsidy Policy	Subsidy level; average fuel price.
5. Which countries are in the top 10 for average petrol price, and how do petrol and diesel compare?	Ranked stacked bar comparison of petrol and diesel prices for the top 10 countries, ranked by average petrol price.	Scenario 2 — Global Fuel Prices & Supply Chain Planning	Country; average petrol price; average diesel price.
6. Which countries are in the bottom 10 for average petrol price, and how do petrol and diesel compare?	Ranked stacked bar comparison of petrol and diesel prices for the bottom 10 countries, ranked by average petrol price.	Scenario 2 — Global Fuel Prices & Supply Chain Planning	Country; average petrol price; average diesel price.
7. How have Brent crude and fuel prices changed over time in the dataset?	Time-series line chart: Brent Crude and Fuel Price Trends. Trends are descriptive and do not establish that crude prices caused fuel-price changes.	Scenario 2 — Global Fuel Prices & Supply Chain Planning	Date; Brent crude (USD/barrel); petrol and diesel prices.
8. How do average petrol prices and tax percentages compare across income levels?	Bar charts: Average Petrol Price by Income Level and Average Tax Percentage by Income Level; the dashboard also includes an Income Level filter for related comparisons.	Scenario 3 — Fuel Prices, Taxation & Economic Development	Income level; average petrol price; average tax percentage.

Dashboard Screenshot 1 — Scenario 1: Fuel Taxation & Subsidy Policy



Evidence image from the supplied dashboard overview. The screenshot shows country tax rankings, regional petrol prices, taxation versus petrol price, and subsidy-level fuel prices.

Dashboard Screenshot 2 — Scenario 2: Global Fuel Prices & Supply Chain Planning



Evidence image from the supplied dashboard overview for the country price comparisons and time-trend analysis.

Dashboard Screenshot 3 — Scenario 3: Fuel Prices, Taxation & Economic Development



Evidence image from the supplied dashboard overview for income-level and subsidy/taxation comparisons.

Interpretation and Scope Note

The workbook analyses the historical CSV dataset available to the project (January 2020–April 2026 as described in the project materials). Values and rankings reflect the dataset and the aggregation used in Tableau. The charts support descriptive comparison and exploration; they do not establish causation, provide live market prices, or replace independent policy/economic research. Rankings in the top/bottom 10 charts are based on average petrol price, not diesel price or a combined fuel score.

Demonstration of Proposed Features

Date	31 October 2026
Team ID	Individual Project (No Team ID)
Project Name	Global Fuel Prices and Taxation Trends: A Comprehensive Global Performance Intelligence Report
Maximum Marks	1 Mark

Demonstration of Proposed Features

This document captures features proposed during project planning and records whether each feature was implemented and can be demonstrated. Statuses are based on the existing Tableau workbook, the latest dashboard screenshots, and the published Tableau Public link. No unverified testing or functionality is claimed.

S.No	Feature Name	Description	Status (Implemented / Partial / Pending)	Demonstrated (Yes / No)	Remarks
1	Scenario 1 — Fuel Taxation & Subsidy Policy	Dashboard combines country fuel-tax comparisons, average petrol price by region, taxation-versuspetrol-price exploration, and subsidy-level comparison.	Implemented	Yes	Four visualisations are present in the latest supplied screenshot. The scatter plot shows association, not causation.
2	Scenario 2 — Global Fuel Prices & Supply Chain Planning	Dashboard compares high- and low-price country groups, petrol and diesel measures, regional average petrol prices, and historical Brent crude/fuel-price trends.	Implemented	Yes	Country groups are ranked by average petrol price. The published workbook includes this dashboard.
3	Scenario 3 — Fuel Prices, Taxation & Economic Development	Dashboard compares average petrol prices and taxes by income level and allows exploration by income, region,	Implemented	Yes	Latest screenshot shows Income Level, Region, and Subsidy Level controls. Interaction speed has not been benchmarked.

		and subsidy level.			
4	Visual comparison	Charts and legends help	Implemented	Yes	Legend meanings vary
	using charts and legends	users compare countries, regions, fuel measures, tax percentages, and subsidy categories.			by dashboard; users should refer to the visible labels.
5	Published Tableau Public workbook	Workbook is published online so viewers can open the dashboards through a browser and navigate between scenario tabs.	Implemented	Yes	Workbook link: https://public.tableau.com/app/profile/harshvardhan.gupta1425/viz/Global_Fuel_Prices_and_Taxation_Trends_17932843176370/Scenario1FuelTaxationSubsidyPolicy? publish=yes
6	Live data refresh / database integration / Tableau Story	Potential extensions include automated data refresh, a separate database connection, and a Tableau Story.	Pending	No	These are not established as implemented in the current CSVbased Tableau project. Do not present them as current features.

Feature Implementation Summary

Total Features Proposed / Tracked	6
Total Features Implemented	5
Total Features Demonstrated	5
Overall Implementation Rate (%)	83.3% (5 of 6 tracked items)

Counting note: the sixth row groups future extensions rather than a single original feature. The 83.3% figure is therefore a simple documentation count for this adapted checklist, not an official platform score or a measured delivery metric.

Evidence and limitations

- Evidence used: latest screenshots of all three dashboards and the published Tableau Public workbook link.
- The dataset was connected from a CSV file directly in Tableau Public; a separate SQL database is not claimed.
- No live data feed, formal user study, load test, mobile responsiveness test, or separate Tableau Story is claimed as completed.
- The dashboards support descriptive comparison and exploration; observed patterns do not establish causation.

Website Development and Integration

1. Overview

As an additional component of the project, a web-based interface was developed to provide convenient access to the project overview and interactive Tableau dashboards. The website complements the existing Tableau Public workbook.

2. Technologies Used

- **Python:** Used as the programming language for the web application.
- **Flask:** Used as the Python web framework to serve the website.
- **HTML:** Used to structure the website content.
- **CSS:** Used to style the website and arrange its visual elements.
- **Tableau Public:** Used to provide the published, interactive data visualisations embedded in the website.
- **GitHub:** Used to store the website source code within the existing project repository.
- **Render:** Used to deploy and host the web application online.

3. Implementation

The website was developed using Flask, with HTML templates and CSS styling. The published Tableau Public dashboard was embedded into the website, allowing visitors to access the project visualisations through a single web interface.

The website source files are organised in the web_integration directory of the existing GitHub repository. The application was deployed through Render and made accessible through a public web address.

4. Live Access

Live Project Website:

https://fuel-prices-taxation-dashboard.onrender.com

GitHub Repository:

https://github.com/Harshvardhan-30/global-fuel-prices-taxation-analysis

Direct Tableau Public Workbook:

https://public.tableau.com/app/profile/harshvardhan.gupta1425/viz/Global_Fuel_Prices_and_Taxation_Trends_17932843176370/Scenario1-FuelTaxationSubsidyPolicy?publish=yes

5. Outcome

The website provides an additional access point for the project, combining its overview with the published Tableau visualisations. This improves the accessibility and presentation of the project while retaining Tableau Public as the platform responsible for the interactive dashboards.